

CS 2050 — Notes 08/31

Compound proposition. Any expression formed from propositional variables using logical operators.

Example. $p \wedge q$; $(p \vee \neg q) \rightarrow (\neg p \wedge \neg q)$

Tautology. A compound proposition that is always true.

Example. $p \rightarrow p$ is a tautology.

Contradiction. A compound proposition that is always false.

Example. $\neg p \wedge p$ is a contradiction.

Contingency. A compound proposition that is neither a tautology nor a contradiction.

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

De Morgan's 1st Law

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

De Morgan's 2nd Law

Conditional-disjunction equivalence: $p \rightarrow q \equiv \neg p \vee q$

Contrapositive law: $p \rightarrow q \equiv \neg q \rightarrow \neg p$

Definition of biconditional: $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

Double negation law: $\neg(\neg p) \equiv p$

Identity Laws

$$p \wedge T \equiv p$$

$$p \vee F \equiv p$$

Domination Laws

$$p \wedge F \equiv F$$

$$p \vee T \equiv T$$

Idempotent Laws

$$p \vee p \equiv p$$

$$p \wedge p \equiv p$$

Commutative Laws

$$p \vee q \equiv q \vee p$$

$$p \wedge q \equiv q \wedge p$$

Associative Laws

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

Distributive Laws

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

Absorption Laws

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

Negation Laws

$$p \vee \neg p \equiv T$$

$$p \wedge \neg p \equiv F$$

Satisfiability. A compound proposition is satisfiable if there is an assignment of truth values to its variables that makes it true.

- Way to show: truth table — a compound proposition is satisfiable if one T is present in the table.
- Logically reducing the statement is possible as well.

Predicate and logic

Propositional logic

- Druids are magic users.
- $2 * 9 = 42$.

Propositional logic failed: $x + y = 27$

Example. $x < 3$: x is the subject, 3 is the predicate. The statement “ $x < 3$ ” can be represented by $P(x)$.

$$P(0) = 0 < 3 = \text{true}, \quad P(4) = 4 < 3 = \text{false}$$

Example. Let $D(x)$ denote “someone in this class likes games.”

$$D(\text{Jade}) \equiv \text{Jade likes games}$$

Multivariable

Example. Let $Q(x, y)$ denote $x = 3 - y$. What are the truth values of $Q(1, 2)$ and $Q(2, 3)$?

$$Q(1, 2) = (1 = 3 - 2) = \text{true}$$

$$Q(2, 3) = (2 = 3 - 3) = \text{false}$$

$\forall$ is the universal quantifier.

$\exists$ is the existential quantifier.

Universe of discourse = domain.

$\exists!$ is the uniqueness quantifier.

Quantifiers over finite domain. If the domain of a quantifier is finite, then a quantified statement can be converted to propositional logic:

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \cdots \wedge P(x_n)$$

$$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \cdots \vee P(x_n)$$